

AI3613 Database System Concepts Homework 2

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Problem 1

Prove the following the characterizations of extraneous attributes:

1. $A \in X$ is extraneous in $X \rightarrow Y$ if and only if $Y \subseteq (X \setminus \{A\})_F^+$.
2. $B \in Y$ is extraneous in $X \rightarrow Y$ if and only if $B \in X_{F'}^+$, where $F' = \{(F \setminus \{X \rightarrow Y\}) \cup \{X \rightarrow (Y \setminus \{B\})\}\}$.

1. if: Suppose $Y \subseteq (X \setminus \{A\})_F^+$. By definition of closure, we have $(X \setminus \{A\}) \rightarrow (X \setminus \{A\})_F^+$. Thus $(X \setminus \{A\}) \rightarrow Y$, and F logically implies $(F \setminus \{X \rightarrow Y\}) \cup \{(X \setminus \{A\}) \rightarrow Y\}$
only if: Suppose $A \in X$ is extraneous in $X \rightarrow Y$, then $(X \setminus \{A\}) \rightarrow Y$. By definition of closure, we have $(X \setminus \{A\}) \rightarrow (X \setminus \{A\})_F^+$, and thus $Y \subseteq (X \setminus \{A\})_F^+$.
2. if: Suppose $B \in X_{F'}^+$, which means on FD F' , $X \rightarrow (Y \setminus \{B\} \cup \{B\})$, or actually $X \rightarrow Y$. This shows F' implies $X \rightarrow Y$. Since F logically implies F' , we have F implies $X \rightarrow Y$ and $X \rightarrow B$, which logically implies F . Thus B is extraneous in $X \rightarrow Y$.
only if: Suppose B is extraneous in $X \rightarrow Y$, then $(F \setminus \{X \rightarrow Y\}) \cup \{X \rightarrow (Y \setminus \{B\})\}$ logically implies F , meaning on F' we have $X \rightarrow Y$, and of course $X \rightarrow \{B\}$. This is $B \in X_{F'}^+$.

Problem 2

Suppose that the following FD's hold on the relation schema $R(A, B, C, D)$:

$$AB \rightarrow C, \quad AB \rightarrow D, \quad C \rightarrow A, \quad D \rightarrow B.$$

1. List all candidate keys of R .
2. Show that R is not in BCNF and give a BCNF decomposition of R .
3. Is your BCNF decomposition dependency preserving? Explain your answer.

1. By observation,

$AB \rightarrow C$, $AB \rightarrow D$, and trivially $AB \rightarrow AB$, so $\{A, B\}$ is a candidate key.

$C \rightarrow A$, $D \rightarrow B$, and trivially $CD \rightarrow CD$, so $\{C, D\}$ is a candidate key.

Similarly, $\{C, B\}$ and $\{D, A\}$ are also candidate keys.

Any single attribute is not a candidate key, and any combination of three attributes is not minimal.

2. For $C \rightarrow A$ in the FD, C is not a superkey, so R is not in BCNF.

Decompose R into $R_1(A, C)$ and $R_2(A, B, D)$, where R_1 contains all attributes in $C \rightarrow A$. In R_2 , $D \rightarrow B$ in FD, D is not a superkey, so R_2 is not in BCNF. Decompose R_2 into $R_3(D, B)$ and $R_4(A, B)$, where R_3 contains all attributes in $D \rightarrow B$. Now R_1, R_3, R_4 are all in BCNF. The decomposition is $R_1(A, C), R_3(D, B), R_4(A, B)$.

3. The decomposition is not dependency preserving. The FDs $AB \rightarrow C$ and $AB \rightarrow D$ are lost in the decomposition because A, B, C or A, B, D never exist in the same relation.

Problem 3

Let F be a set of FD's holds on a schema R and $R_1, \dots, R_n$ be a decomposition of R . Furthermore, assume the following:

- For every $X \rightarrow Y$ in F , there exists some R_i that contains all the attributes in XY .
- At least one schema in the decomposition contains a candidate key of R .

Prove that the decomposition $R_1, \dots, R_n$ is join lossless. With this property, we can show that any decomposition produced by the 3NF synthesis algorithm is join lossless.

First of all, according to the first assumption, for any FD $X \rightarrow Y$ in F , there exists some R_i that contains all the attributes in XY . That means, none of the FDs in F is lost in the decomposition. This is $\bigcup_{i=1}^n F_i = F$. However, the closures of the FDs are not necessarily the same for F_i is restricted to R_i .

Secondly, at least one schema in the decomposition contains a candidate key of R . This shows there exists at least one R_c that contains a candidate key of R , which can uniquely determine all the attributes in R for any certain record. Now suppose we construct a template table with all the attributes in R based on R_c , we can locate the position of the contents in any table with R_i by the candidate key in R_c . This shows the closure of $\bigcup_{i=1}^n F_i$ is the same as F^+ . Thus the decomposition is join lossless.

Maybe I fail to explain it clearly. Please inform me if there's crucial error or ambiguity.

Problem 4

How long does it take you to finish the assignment? Give a score (1,2,3,4,5) to the difficulty of each problem. List all your collaborators if you have any.

Haven't seen this problem since *Algorithm Design and Analysis* course. The times I completed this assignment were very *scattered* but I guess it only took hours including the time I review the required knowledge. If my answer is correct, I would rate it 3, 2, 4 for each problem. If there's any crucial mistake in my answer, I would rate it 5. Finished independently.